Number Theory and Graph Theory

Chapter 4

Primitive roots and Quadratic residues

By

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Module-1: Primitive roots

Objectives

- Group of units of $\mathbb{Z}_n(=\{0,1,2,\ldots,n-1\})$.
- Order of an integer Modulo n, where $n \in \mathbb{N}$.
- Primitive roots of n.

Let $n \in N$. We write $U_n = \{x \in \mathbb{N} | 1 \leq x \leq n, (x,n) = 1\}$. So, U_n **is set of all numbers from 1 to** n **that are relatively prime to** n. For example, $U_8 = \{1,3,5,7\}$, $U_{12} = \{1,5,7,11\}, U_p = \{1,2,3,\ldots,p-1\}$.

Why are we interested in these numbers?

- As we know $\mathbb{Z}_n = \{0,1,2,\ldots,n-1\}$ forms a group with respect to addition modulo n. **But, in general,** $\mathbb{Z}_n^* = \mathbb{Z}_n \setminus \{0\}$ **does not form a group with respect to multiplication modulo** n**.**
- For example $\mathbb{Z}_6 = \{0,1,2,3,4,5\}$ is a group with respect to addition modulo 6, but $\mathbb{Z}_6^* = \mathbb{Z}_6 \setminus \{0\} = \{1,2,3,4,5\}$ is not a group with respect to multiplication modulo 6 as $2 * 3 = 6 \equiv 0 \pmod 6$. Also, the elements $2,3$ and 4 do not have multiplicative inverses modulo 6. Or equivalently,

  $\boxed{ax \equiv 1 \pmod 6 \text{ does not have a solution whenever } a = 2,3,4.}$
- Next obvious question is "Is $U_n = \{x \in \mathbb{N} | 1 \leq x \leq n, (x,n) = 1\}$ a group with respect to multiplication modulo n."

  Answer: YES. Verify it.

- The next question is "Is $U_n = \{x \in \mathbb{N} | 1 \leq x \leq n, (x,n) = 1\}$ cyclic for all $n \in \mathbb{N}$"?.
  Answer: NO. For example, U_8 is not cyclic. In general, U_n is cyclic whenever $n = 2, 4, p^k, 2p^k \;\; k \geq 1$, where p is an odd prime. For $n = 2, 4, p^k, 2p^k \;\; k \geq 1$, the generators of U_n are called **primitive roots modulo n.**

- For example, for $n = 5$, $U_5 = \{1,2,3,4\}$ and $2,3$ are primitive roots modulo 5. As
$$U_5 = \{2^0, 2^1 \equiv 2 \pmod 5, 2^2 \equiv 4 \pmod 5, 2^3 \equiv 3 \pmod 5\}.$$
  Verify that 3 is also a primitive root modulo 5. In other words 2 and 3 are generators of the cyclic group U_5.

- Further, U_n is the set of all generators of the additive cyclic group $\mathbb{Z}_n$, addition modulo n. That is, U_n is the collection of all elements in $\mathbb{Z}_n$ that have multiplicative inverses. Hence, U_n is called the group of units of $\mathbb{Z}_n$.

The Order of an Integer

Fix a positive integer n. Then, by Euler's theorem, for any $a \in \mathbb{N}, \gcd(a,n) = 1$, one has $a^{\varphi(n)} \equiv 1 \pmod n$. Therefore, there exists a smallest positive integer, say x, such that $a^x \equiv 1 \pmod n$ and $a^k \not\equiv 1 \pmod n$, for $1 \leq k \leq x-1$.

Definition 1. *Fix a positive integer n and let $a \in \mathbb{N}$ with $\gcd(a,n) = 1$.*

1. *Then, the least positive integer x such that $a^x \equiv 1 \pmod n$ is called the order of a modulo n, denoted $ord_n(a)$.*
2. *If $ord_n(a) = \varphi(n)$ then, a is said to be a* **primitive root modulo** *n.*

For example, for $n = 13$,
$$3^1 \equiv 3 \pmod{13}, 3^2 \equiv 9 \pmod{13}, 3^3 \equiv 1 \pmod{13}.$$

So $ord_{13}(3) = 3$, $12^2 \equiv 1 \pmod{13}$ so $ord_{13}(12) = 2$. (In fact $n-1 \equiv -1 \pmod{n}$ for every n and hence $ord_n(n-1) = 2$.) Whereas, $ord_{13}(2) = 12 = \varphi(13)$, so 2 is a primitive root modulo 13.

Problem 2. 1. *Show that there are no primitive roots modulo* 8.

Solution:$U_8 = \{1,3,5,7\}$ *it is easy to check that* $3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$. *Hence*

$$ord_8(a) \leq 2$$

for all $a \in U_8$.

2. *Show that there are no primitive roots modulo* 16.

Solution:$U_{16} = \{1,3,5,7,9,11,13,15\}$. *So* $\varphi(16) = |U_{16}| = 8$. *Verify that*

$$a^4 \equiv 1 \pmod{16}, \text{ for all } a \in U_{16}.$$

Hence, there is no element a with $(a,16) = 1$ *such that* $ord_{16}(a) = 8 = \varphi(16)$.

3. *Fix a positive integer n and let* $a \in \mathbb{N}$ *such that* $\gcd(a,n) = 1$. *If a is primitive root of n and* $U_n = \{a_1, a_2, \ldots, a_{\varphi(n)}\}$ *then,*

$$\{a_1, a_2, \ldots, a_{\varphi(n)}\} \equiv \{a, a^2, \ldots, a^{\varphi(n)}\} \pmod{n}.$$

Solution:As a is a primitive root modulo n, the numbers $a^i \pmod{n}$ *and* $a^j \pmod{n}$ *are distinct, whenever* $1 \leq i \neq j \leq \varphi(n) - 1$. *Moreover,* $\gcd(a,n) = 1$ *implies that* $\gcd(a^k, n) = 1$ *for all* $1 \leq k \leq \varphi(n)$. *Thus, the required result follows.*

4. *Fix a positive integer n and assume that n has a primitive root. Then, the number of primitive roots of n equals* $\varphi(\varphi(n))$.

Solution:Note that if a is a primitive root modulo n then, $ord_n(a) = \varphi(n)$. *Now, observe that for* $1 \leq k \leq \varphi(n)$, $ord_n(a^k) = \varphi(n)$ *if and only if* $\gcd(k, \varphi(n)) = 1$. *But, by definition*

$$|\{k : 1 \leq k \leq \varphi(n), \gcd(k, \varphi(n)) = 1\}| = \varphi(\varphi(n))$$

and hence the required result follows.

5. *Find all primitive roots modulo* 13.

   Solution:Using the example given before Problem 2, we see that 2 *is a primitive root modulo* 13*. As* $\varphi(13) = 12$ *and the number* $1, 5, 7$ *and* 11 *are coprime to* 12*, we see that* $2 = 2^1, 6 \equiv 2^5 \pmod{13}, 11 \equiv 2^7 \pmod{13}$ *and* $7 \equiv 2^{11} \pmod{13}$ *are the primitive roots modulo* 13*. Thus, the number of primitive roots modulo* 13 *equals* $4 = \varphi(\varphi(13)) = \varphi(12)$.

Theorem 3. *Let* $n = 2^k$*, for some positive integer* k*. Then,* n *has a primitive root modulo* n *(*U_n *is cyclic) whenever* $k = 1$ *or* $k = 2$*.*

Proof. If $k = 1$ or $k = 2$ then, $U_2 = \{1\}$ and $U_4 = \{1, 3\}$ are indeed cyclic. So, we now show that U_{2^k} is not cyclic for $k \geq 3$. To do so, it is sufficient to prove that $ord_{2^k}(x) < \varphi(2^k) = 2^{k-1}$, for all $x \in U_{2^k}$. In fact, we will use induction to show that $x^{2^{k-2}} \equiv 1 \pmod{2^k}$, for all $x \in U_{2^k}$ whenever $k \geq 3$.

Base case $k = 3$ In Problem 2.1 we showed that U_8 is not cyclic and hence has no primitive root modulo 8.

Induction hypothesis Suppose the result is true for k, where $k \geq 3$. That is, we assume that $x^{2^{k-2}} \equiv 1 \pmod{2^k}$, for all $x \in U_{2^k}$ with $k \geq 3$. Or equivalently, $x^{2^{k-2}} = 1 + m2^k$ for some positive integer m. We will now prove the result for $k+1$. That is, we need to show that for every $y \in U_{2^{k+1}}$, $y^{2^{k-1}} \equiv 1 \pmod{2^{k+1}}$.

So, let $y \in U_{2^{k+1}}$. Then, either $1 \leq y < 2^k$ or $y = x + 2^k$ with $1 \leq x < 2^k$. In either case, $y = x + t2^k$, where $t \in \{0, 1\}$ and $1 \leq x < 2^k$. Hence, using $k \geq 3$ and the binomial theorem, we have

$$y^{2^{k-1}} = (x + t2^k)^{2^{k-1}} \equiv x^{2^{k-1}} \pmod{2^{k+1}} \equiv (x^{2^{k-2}})^2 \pmod{2^{k+1}}.$$

Thus, using $x^{2^{k-2}} = 1 + m2^k$ for some positive integer m and $k \geq 3$, we get

$$y^{2^{k-1}} \equiv (x^{2^{k-2}})^2 \pmod{2^{k+1}} \equiv (1 + m2^k)^2 = 1 + m2^{k+1} + m^2 2^{2k} \equiv 1 \pmod{2^{k+1}}.$$

□

We will now show that U_{p^k} is cyclic for all odd primes p and for all $k \geq 1$. To do so, we need the following results.

Theorem 4 (Division algorithm)**.** *Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial of degree $n \geq 1$ with integer coefficients. If $g(x) = x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$ is a non-zero polynomial then there exists unique polynomials $q(x)$ and $r(x)$ with integer coefficients such that $f(x) = g(x)q(x) + r(x)$ with either $r(x)$ is identically the zero polynomial or* $\deg(r(x)) < \deg(g(x)) = m$.

Proof. We first proof the existence of the polynomials $q(x)$ and $r(x)$.
The result is clearly true if $\deg(f(x)) < \deg(g(x))$ as in this case, $f(x) = 0g(x) + f(x)$.

So, let the result be true for all polynomials of degree $n-1$ and consider the polynomial $f(x)$ of degree n. If $n < m$, we are done. So, let us assume that $n \geq m$. Then, we see that the polynomial $h(x) = f(x) - a_n x^{n-m} g(x)$ has integer coefficients and $\deg(h(x)) \leq n-1$. Thus, be induction hypothesis, there exists polynomials $q_1(x)$ and $r_1(x)$ with integer coefficients such that

$$f(x) - a_n x^{n-m} g(x) = h(x) = g(x)q_1(x) + r_1(x)$$

with either $r_1(x)$ as the zero polynomial of $\deg(r_1(x)) < \deg(g(x))$. Hence, $f(x) = g(x)\left(a_n x^{n-m} + q_1(x)\right) + r_1(x)$. Thus, we obtained the required result as we have obtained $q(x) = a_n x^{n-m} + q_1(x)$ and $r(x) = r_1(x)$.

The proof of uniqueness is left for the reader. □

Lemma 5. *Let $f(x)$ be a polynomial of degree $n \geq 1$ with integer coefficients. Then, $f(a)$ is the remainder when $f(x)$ is divided by $x - a$.*

Proof. The division algorithm applied to the polynomials $f(x)$ and $x-a$ gives us unique polynomials $q(x)$ and $r(x)$ such that $f(x) = (x-a)q(x) + r(x)$, with either $r(x)$ as the zero polynomial or $\deg(r(x)) < \deg(x-a) = 1$. Hence, in either case, $r(x)$ is a constant. Moreover, we see that

$f(a) = (a-a)q(a) + r(a) = r(a)$. So, $r(x)$ is a constant implies that $r(x) = r(a) = f(a)$ for all x. Hence, $f(a)$ is the remainder when $f(x)$ is divided by $x-a$. □

Corollary 6. *Let $f(x)$ be a polynomial of degree $n \geq 1$ with integer coefficients. Then, a is a root of $f(x)$ if and only if $x-a$ is a factor of $f(x)$.*

Proof. By the division algorithm, there exists $q(x)$ and $r(x)$ such that $f(x) = (x-a)q(x) + r(x)$, with either $r(x)$ as the zero polynomial or $\deg(r(x)) < \deg(x-a) = 1$.

If a is a root of $f(x)$ then, $f(a) = 0$ and hence by Lemma 5, we see that $f(x) = (x-a)g(x)$.

Conversely, if $x-a$ is a factor of $f(x)$ then $r(x)$ is the zero polynomial and hence $f(a) = 0$. Thus, the required results follow.

□

Theorem 7 (Lagrange's theorem)**.** *Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ be a polynomial of degree $n \geq 1$, with integer coefficients. Let p be a prime number such that $p \nmid a_n$. Then $f(x)$ has at most n incongruent roots, counted with multiplicity, modulo p.*

Proof. We prove the result by induction on n, the degree of the polynomial.
For $n = 1$, $f(x) = a_1 x + a_0$ with $p \nmid a_1$. Then, using Theorem 2 of Module 5 of Chapter 2, we know that the linear congruence $a_1 x + a_0 \equiv 0 \pmod{p}$ has exactly one solution modulo p.

Suppose the theorem is true for all polynomials of degree less than or equal to $n-1$ and let $f(x)$ be a polynomial of degree n. If $f(x)$ has no root modulo p, we are done. So, let us assume that α_0 is an incongruent root of $f(x)$ of multiplicity k modulo p. Then, using Corollary 6 repeatedly, we see that $f(x) \equiv (x-\alpha_0)^k g(x) \pmod{p}$ for some polynomial $g(x)$. Since the leading coefficient of $g(x)$ and $f(x)$ are the same and $\deg(g(x)) \leq n-1$, by induction hypothesis $g(x)$ has at most $\deg(g(x)) = n-k$ roots, counted with multiplicity, modulo p. Moreover,

1. $g(\alpha_0) \neq 0$ as α_0 is a root of $f(x)$ of multiplicity k modulo p,

2. β is a root of $g(x)$ modulo p if and only if β is a root of $f(x)$ modulo p as $\beta \neq \alpha_0$ and

$$f(\beta) = (\beta - \alpha_0)^k g(\beta).$$

Thus, $f(x)$ has at most $n = k + (n-k)$ incongruent roots, counted with multiplicity, modulo p. $\square$

Theorem 8. *Let p be a prime and $d|p-1$. Then $x^d - 1$ has exactly d incongruent solutions modulo p.*

Proof. Since $d|p-1$ there exists $t \in \mathbb{Z}$ such that $p-1 = td$. Hence,

$$x^{p-1} - 1 = x^{td} - 1 = (x^d - 1)(x^{d(t-1)} + x^{d(t-2)} + x^{d(t-3)} + \cdots + x + 1).$$

Let $f(x) = x^{d(t-1)} + x^{d(t-2)} + x^{d(t-3)} + \cdots + x + 1$. Then, $f(x)$ is a polynomial with integer coefficients and of degree $dt - d = p - 1 - d$. Moreover, p does not divide its leading coefficient and hence by Lagrange's Theorem 7 $f(x)$ has at most $p-1-d$ incongruent roots modulo p. But, from Fermat's Little theorem, we know that the polynomial $x^{p-1} - 1$ has exactly $p-1$ incongruent solutions modulo p (the roots are $1, 2, 3, \ldots, p-1$). Hence, $x^d - 1$ and $f(x)$ have exactly respectively, d and $p-1-d$, incongruent solutions modulo p. Thus, the required result follows. $\square$

Theorem 9. *Let p be an odd prime. Then, U_p is cyclic.*

Proof. In fact we claim that for each $d|p-1$ there are exactly $\varphi(d)$ elements in U_p of order d. This will give the required result as for $d = p-1$, we will have $\varphi(p-1)$ elements in U_p or order $p-1$.

To prove the claim, we consider the set $\Gamma(d) = \{x \in U_p | ord_p(x) = d\}$ and prove that $|\Gamma(d)| = \varphi(d)$.

Note that the sets $\Gamma(d)$ partition the set U_p and hence $p - 1 = \sum\limits_{d|p-1} \gamma(d)$. Also, from Theorem 6 of Module 2 of Chapter 3 we have $p - 1 = \sum\limits_{d|p-1} \varphi(d)$. Thus, we have $\sum\limits_{d|p-1} (\varphi(d) - \gamma(d)) = 0$. Hence, it is sufficient to show that $\varphi(d) - \gamma(d) \geq 0$, for all $d|p-1$.

Note that the result is trivially true whenever $\Gamma(d) = \emptyset$. Hence, let us assume that $\Gamma(d) \neq \emptyset$ and let $a \in \Gamma(d)$. Then $1, a, a^2, \ldots, a^{d-1}$ are all distinct and incongruent solutions of the polynomial $x^d - 1$. But from Theorem 8 $x^d - 1$ has exactly d incongruent solutions and hence

$$\Gamma(d) \subseteq \{1, a, a^2, \ldots a^{d-1}\}.$$

Now, let $c \in \Gamma(d)$. Then, $c = a^i$ for some $i, 1 \leq i \leq d-1$. We claim that $\gcd(i,d) = 1$. For if, $\gcd(i,d) = j > 1$ then, $c^{d/j} = a^{id/j} = 1$ and hence $ord_p(c) < d$, a contradiction of $ord_p(c) = d$. Thus, $\Gamma(d) \subseteq \{a^k | (k,d) = 1, 1 \leq k \leq d\}$. □

For the following result we provide the sketch of the proof. Complete proof is left as an exercise.

Theorem 10. *Let p be an odd prime. Then, U_{p^k} is a cyclic group for all $k \geq 1$.*

Proof.

1. Since U_p is cyclic, there exists a generator for U_p. Let a be one such generator.
2. Then, we show that either a or $a+p$ is a generator for U_{p^2}.
3. Finally, we show that if b is generator for U_{p^2} then, b is generator for U_{p^k} for all $k \geq 2$.

□